Testing Whether Attention Recovers Finite-Difference Stencils in PDE Transformers

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Abstract

We test whether attention weights recover a known finite-difference update in a transformer. Before attention is used to support scientific discovery, it should recover a known mechanism in a controlled setting. We study encoder-only transformers trained to predict the next state of heat diffusion and Lax-Friedrichs advection data generated by finite-difference schemes. The models predict accurately, but their attention weights neither match the corresponding stencils nor follow changes in the stencil coefficients. The same analysis recovers the stencil in attention-only controls, where replacing the learned weights with the exact stencil leaves prediction error effectively unchanged. In the trained transformers, fixing attention increases prediction error, while replacing it with the stencil causes a much larger increase. Fourier comparisons show condition-dependent agreement on frequencies present in the generated states, but the transformer response is closer to copying the current state at the tested higher frequencies. Attention therefore contributes to prediction without directly reproducing the finite-difference update, so its weights alone do not explain how the transformer performs the computation.

1 Introduction & Related Work

Neural operators learn mappings between functions and can approximate solution operators for partial differential equations (PDEs) [Lu et al., 2021, Li et al., 2021]. Attention-based operator learners have also been connected to Galerkin-style approximations [Cao, 2021]. These results show predictive capability, but not whether a trained model's internal weights recover a numerical method. The finite-difference stencil provides a clear test of whether attention recovers its coefficients as the PDE parameters change. Because each token represents one grid point, attention combines value vectors across spatial positions [Vaswani et al., 2017]. Attention concentrated on neighboring grid points may therefore represent a finite-difference stencil. Studies of NLP models show that attention weights can disagree with gradient-based measures of input importance and that substantial changes to the attention weights do not always change the predictions [Jain and Wallace, 2019, Serrano and Smith, 2019]. Other work argues that claims about attention should be tested against explicit baselines and by altering the attention weights. [Wiegreffe and Pinter, 2019]. Methods that learn finite-difference coefficients from data can remain interpretable because the form of the numerical update is specified in advance [Bar-Sinai et al., 2019]. Nonlocal Attention Operator and Neural Interpretable PDEs use specialized architectures for interpretable physics discovery, learning an attention-based kernel map and a Fourier-space convolution kernel, respectively [Yu et al., 2024, Liu and Yu, 2025]. In contrast, we test whether attention in a standard transformer recovers a known finite-difference update and examine the full model separately because the computation may be spread across several components. We test whether attention weights at offsets $-1, 0, 1$ match the finite-difference coefficients and track their changes with r or ν. Heat changes the center coefficient, whereas Lax-Friedrichs changes the left-right balance while keeping the center coefficient zero.

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2 Methods

We study two finite-difference schemes on a periodic one-dimensional grid. For the heat equation $\partial_t u = \alpha \partial_{xx} u$, the update is

$$u_i^{t+1} = r u_{i-1}^t + (1-2r) u_i^t + r u_{i+1}^t, \qquad s_{\mathrm{Heat}}(r) = [r,\, 1-2r,\, r]\,. \tag{1}$$

where $r = \alpha \Delta t / \Delta x^2$ is the diffusion number [LeVeque, 2007, Courant et al., 1967]. We use $r \in \{0.1, 0.25, 0.4\}$, all within the stability bound $r \leq 0.5$. These values change the center weight while preserving symmetry.

For linear advection, the Lax-Friedrichs stencil for offsets $(-1, 0, 1)$ is

$$s_{\mathrm{LF}}(\nu) = \left[\frac{1+\nu}{2},\, 0,\, \frac{1-\nu}{2}\right], \tag{2}$$

where $\nu = a\Delta t / \Delta x$ is the Courant number [LeVeque, 2007, Courant et al., 1967]. We use $\nu \in \{0.1, 0.25, 0.4\}$. all within the stability bound $|\nu| \leq 1$. The center weight is zero, and the difference between the left and right weights grows with ν. The two systems therefore test whether attention follows the changing center weight for heat and the changing left-right weights for Lax-Friedrichs.

On a periodic grid, both updates define circulant linear operators with exact Fourier representations [Gray, 2006, Trefethen, 2000]. We use their stencils and Fourier multipliers as references in our analyses. Appendix A, derives the updates, and Appendix A.3 gives the exact reference expressions.

2.1 Model and Experimental Design

We use an encoder-only transformer based on Vaswani et al. [2017] with sinusoidal positional encoding and Pre-LayerNorm [Xiong et al., 2020] implemented in PyTorch [Paszke et al., 2019]. Each of the $N_x = 64$ grid points is one token. The model has four layers, four attention heads per layer, $d_{\mathrm{model}} = 64$, $d_{\mathrm{ff}} = 256$, and dropout 0.1, for 200,129 total parameters. It uses global self-attention and contains no physics-informed loss, local attention mask, or imposed numerical operator.

The two equations and three parameter values define six experimental conditions. For each condition, we train 20 independent initialized models and evaluate 30 untrained models with independent random initializations. This gives 120 trained models and 180 random baselines. Parameter choices and stencil coefficients are listed in Appendix B.

Models minimize single-step MSE using Adam [Kingma and Ba, 2015]. We use a batch size of 256, clip the gradient norm at 1, and cosine-anneal the learning rate of 10^{-3}, to 10^{-5} over each run. We average over samples and spatial positions. Attention collection, epoch selection, and convergence criteria are provided in Appendices C, D, and D.1.

2.2 Data Generation

We generate data from the finite-difference schemes on a periodic grid with $N_x = 64$, $\Delta x = 1/64$, $\Delta t = 10^{-4}$, and $N_t = 2000$ time steps. Each condition contains 800 training, 100 validation, and 100 test trajectories. Trajectories are assigned to only one split, so no initial condition appears in more than one set. Models are trained for single-step prediction of u^{t+1} from u^t using consecutive state pairs.

Initial conditions are random superpositions of Fourier modes $k = 1, \ldots, 5$, with independently sampled amplitudes and phases, followed by normalized $[-1, 1]$. The same initial condition distribution is used across both PDEs and all parameter values; further details are provided in (Appendix E). The training data contain only Fourier modes $k = 1, \ldots, 5$. We therefore report Fourier results for these modes separately from frequencies not seen during training.

3 Analysis Metrics

We separately test whether training changes attention, learned attention matches the known finite-difference stencil, input-dependent attention changes affect predictions, and the model and finite-difference update respond similarly to the same input change.

Attention by relative offset. For head h in layer ℓ, let $A^{(\ell,h)} \in \mathbb{R}^{N_x \times N_x}$ be its attention matrix averaged over held-out test inputs. On the periodic grid, define

$$p^{(\ell,h)}(d) = \frac{1}{N_x} \sum_{i=0}^{N_x-1} A^{(\ell,h)}_{i,(i+d) \bmod N_x}. \tag{3}$$

Here, $p^{(\ell,h)}(d)$ is the mean attention at offset d. We report row entropy, normalized entropy across offsets [Shannon, 1948], attention mass within $|d| \leq 1$ and $|d| \leq 4$, and signed and absolute Pearson correlation with the three stencil coefficients [Pearson, 1895]. Correlation measures relative weighting, not coefficient magnitude. Appendix F defines these metrics.

Trained and random attention. For each condition, we compare 20 trained models with 30 random initializations. We average each metric over all 16 heads before testing, so each model contributes one value. We use exact two-sided Mann-Whitney tests, Cliff's δ, and Benjamini-Hochberg correction across the six conditions separately for each metric [Mann and Whitney, 1947, Cliff, 1993, Benjamini and Hochberg, 1995]. For each head, we use Jensen-Shannon divergence to compare normalized attention within $|d| \leq 4$ with the stencil, and stencil weights reconstructed from low Fourier modes [**?**]. We count a head as a match only when both concentration and agreement exceed thresholds set by random initialization. Appendix G gives the full procedure.

Parameter tracking. For each head, we test two relative-offset measures that vary with the stencil parameters.

$$C(p) = \frac{p(0)}{p(-1) + p(0) + p(1)}, \qquad A(p) = \frac{p(1) - p(-1)}{p(-1) + p(1)}. \tag{4}$$

For heat, the expected value is $C_{\text{heat}}(r) = 1 - 2r$. This gives 0.8, 0.5, and 0.2 for $r = 0.1$, 0.25, and 0.4. For Lax-Friedrichs, the expected value is $A_{\text{LF}}(\nu) = -\nu$ under our source-offset convention. We fit $Y = \alpha + \beta X$, where X is the expected value and Y is the measured value. We include all heads. A perfect match gives $\alpha = 0$ and $\beta = 1$. Confidence intervals use 2,000 bootstrap samples of complete model seeds within each condition. We keep all 16 heads from each selected checkpoint [Field and Welsh, 2007]. Appendix H gives the full fitting and averaging procedure.

Attention substitutions. For each checkpoint, we replace attention in three ways. We use each head's mean over the validation data, one validation mean shared by all heads in a layer, or a matrix built from the three finite-difference coefficients with rows that sum to one. We make each replacement after the attention softmax and before attention is applied to the value vectors. We evaluate it on separate test data. The first removes changes across inputs. The second also removes differences among heads in a layer. The third replaces learned attention with the finite-difference stencil. We leave the trained value and output projections, residual paths, and feed-forward layers unchanged. As a baseline, we copy the current state, $\widehat{u}^{t+1} = u^t$, and denote its test error by MSE_{copy}. We measure the error increase as

$$E_{\text{replacement}} = \frac{\text{MSE}_{\text{replacement}} - \text{MSE}_{\text{original}}}{\max(\text{MSE}_{\text{copy}} - \text{MSE}_{\text{original}}, 10^{-12})}. \tag{5}$$

A value of zero means that the replacement does not change test MSE. A value of one means that it raises test MSE to the copy baseline. We compare test MSEs within each model and bootstrap complete model seeds. Appendix I explains how the matrices and errors are computed and gives the checks and full results.

Attention Only Control. We fit one control for each condition and data seed, giving 120 controls in total. Each control learns three nonnegative weights that sum to one. The weights apply to the left neighbor, the current point, and the right neighbor: $\widehat{u}_i^{t+1} = \sum_{d \in \{-1,0,1\}} w_d u_{i+d}^t$. This weighted sum is the full model. The finite-difference weights are not provided during training.

Full-model Jacobian and Fourier comparison. For each checkpoint, we compute the input-output Jacobian J for 16 held-out states and average the results. The residual connection adds the identity I. We therefore study the learned update $K = J - I$, then measure the squared difference between K and its closest circulant matrix. We Fourier transform the closest circulant matrix to J. We compare its complex values with those from the one-step PDE update and the identity. For each model, we

define $\Delta = \mathrm{RMSE}_I - \mathrm{RMSE}_{\mathrm{PDE}}$. A positive Δ means that the model is closer to the PDE update. The generated states contain only modes $k = 1, \ldots, 5$. We measure error once using the observed power of each mode [**?**] and once by weighting all five modes equally. We report unseen modes $k = 6, \ldots, 8$ and $k = 9, \ldots, 32$ separately. Confidence intervals resample complete model seeds. Appendix J gives the definitions, checks, and results for each condition.

4 Results

Relative to copying the current state, $\widehat{u}^{t+1} = u^t$, the trained models reduce test MSE by 99.6–99.8% for heat and by more than 99.8% for Lax-Friedrichs.

Attention changes, but not consistently toward the stencil. Training lowers mean row entropy in every condition. Every trained model has lower entropy than every random model (Cliff's $\delta = -1$). All six differences remain significant after BH correction. Other measures do not consistently move toward the stencil. Absolute correlation with the stencil increases for heat but decreases for Lax-Friedrichs. Attention mass within $|d| \leq 1$ decreases in five of six conditions. No trained head meets the stencil-agreement criteria ($0/1{,}920$). No held-out random head meets them either. Changes in r and ν also do not produce the expected changes. For heat, measured centrality remains near 0.335, while the expected value falls from 0.8 to 0.2 ($\beta = 0.0002$, 95% CI $-0.0039, 0.0044$). For Lax-Friedrichs, measured asymmetry increases from 0.008 to 0.143, while the expected value decreases from -0.1 to -0.4 ($\beta = -0.450$, 95% CI $-0.682, -0.217$). All 120 attention-only controls meet the stencil-agreement criteria. Their weights match the stencil coefficients with a maximum absolute error of 1.25×10^{-3}. The tracking slope is 1.000 for heat and 1.001 for Lax-Friedrichs. The second slope differs from one by about 0.1% (Figure 1a).

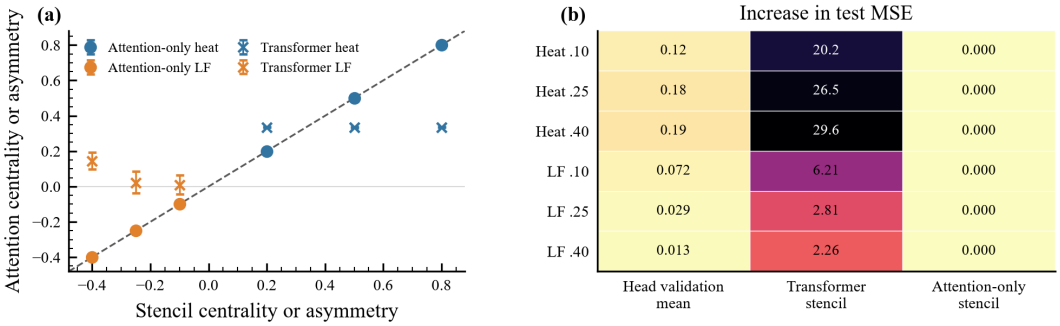


Figure 1: (a) Finite-difference stencil centrality or asymmetry versus attention centrality or asymmetry. Markers show 20-seed condition means, bars show 95% bootstrap intervals, and the dashed line, $y = x$, for agreement. (b) Test-MSE increase after fixing each head at its validation-set mean or replacing attention with the stencil. Cells show 20-seed condition means.

Fixing attention degrades prediction. Fixing each trained head at its validation-set mean increases test MSE in all 120 transformer models. Across the 20 models in each condition, the mean normalized increase ranges from 0.013 to 0.195. Replacing the learned attention matrices with the corresponding finite-difference stencil produces larger increases, ranging from 2.26 to 29.64. Because all six condition means are greater than one, the models with stencil replacement are, on average, less accurate than simply copying the current state. In the attention-only controls, replacing the learned attention with the exact stencil changes the normalized test MSE by at most 1.12×10^{-7} (Figure 1b). The stencil replacement therefore leaves the control predictions effectively unchanged but does not preserve the predictions of the trained transformers. The transformer may therefore produce accurate predictions without learning the finite-difference stencil in its attention matrices. This does not rule out stencil-related computation elsewhere in the network.

Fourier agreement depends on frequency and weighting. The generated states use Fourier modes $k = 1$-5. On these modes, using weights derived from the generated states makes the PDE response closer than the prediction $\widehat{u}^{t+1} = u^t$ in four of six conditions; heat $r = 0.25$ is inconclusive, and

the prediction is closer for heat $r = 0.4$. With equal weights, the PDE response is closer in two conditions, the prediction is closer in three, and one is inconclusive (Figure 2). For both $k = 6$-8 and $k = 9$-32, the prediction is closer in all six conditions, and all 12 model-seed bootstrap intervals lie below zero. Fourier agreement therefore depends on the condition and weighting within the generated frequencies and does not extend beyond them under this comparison. The results do not determine why the transformer learns this frequency-dependent response; the restricted training spectrum is one possible explanation because the loss does not constrain unseen frequencies.

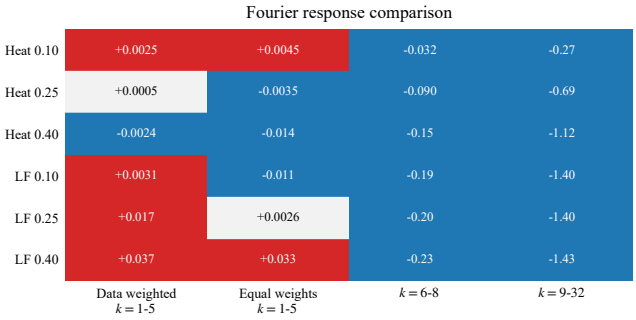


Figure 2: Each entry is the mean across model seeds of $\Delta = \mathrm{RMSE}_{\mathrm{copy}} - \mathrm{RMSE}_{\mathrm{FD}}$. Positive values mean that the finite-difference response is closer to the transformer response. Negative values mean that the copy-current-state response is closer. Red marks 95% intervals above zero, blue marks intervals below zero, and gray marks intervals containing zero. Data weighting uses the power of each mode in the generated training states. Equal weighting gives modes $k = 1, \ldots, 5$ the same weight.

5 Discussion

Training lowers attention entropy, but no trained head matches the finite-difference stencil. The same analysis identifies all 120 attention-only controls, follows their parameter changes, and shows that replacing their attention with the finite-difference stencil leaves test MSE effectively unchanged. The trained-transformer result is therefore not simply a failure of the analysis. The transformer may make accurate predictions without learning the stencil in its attention matrices, although stencil-related computation may occur elsewhere in the network.

Fourier agreement varies across conditions and weightings for the training modes $k = 1$-5. For the unseen modes $k = 6$-8 and $k = 9$-32, the transformer response is closer to copying the current state than to the finite-difference update in every condition. The cause is unknown, and the limited training frequencies are one possible explanation. Because the replacements change only attention, they do not determine what the other trained components contribute.

Scope and limitations. We study one encoder-only transformer, two one-dimensional linear schemes, one grid, one-step prediction, and initial conditions with five Fourier modes. The attention-only control shows that the analysis finds the stencil when attention is the only spatial mixing operation. It does not show whether the same update can be identified when computation is spread across attention and other parts of the transformer. Our conclusions apply only to the architecture, equations, data, and prediction setting studied here.

Responsible Use Statement. This work uses synthetic PDE data and involves no people, personal data, or deployed decision systems. The main risk is over-interpreting attention or generalizing beyond two one-dimensional equations and one architecture. We address this by testing attention replacements and complete-model responses and limiting our claims to the evaluated settings.

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A Stencil Derivations

We derive the two explicit finite-difference updates used to generate the data.

A.1 Derivation of the Heat Diffusion Stencil

Discretize the time derivative with a forward difference:

$$\frac{\partial u}{\partial t} \approx \frac{u_i^{t+1} - u_i^t}{\Delta t},$$

and the spatial derivative with the second-order central difference LeVeque [2007]:

$$\alpha \frac{\partial^2 u}{\partial x^2} \approx \alpha \frac{u_{i-1}^t - 2u_i^t + u_{i+1}^t}{\Delta x^2}$$

Setting these equal and multiplying by Δt gives

$$u_i^{t+1} - u_i^t = \frac{\alpha \Delta t}{\Delta x^2}(u_{i-1}^t - 2u_i^t + u_{i+1}^t)$$

Define the dimensionless heat parameter $r = \alpha \Delta t / \Delta x^2$. For the forward-time centered-space update, stability requires $r \leq 1/2$ Courant et al. [1967]. Rearranging yields the update rule

$$u_i^{t+1} = r \cdot u_{i-1}^t + (1 - 2r) \cdot u_i^t + r \cdot u_{i+1}^t.$$

The coefficients $[r,\ 1-2r,\ r]$ form the finite difference stencil against which we compare attention weights throughout this paper.

A.2 Derivation of the Advection Equation

We consider the linear advection equation

$$\frac{\partial u}{\partial t} + v \frac{\partial u}{\partial x} = 0, \tag{6}$$

with constant advection speed $v > 0$, discretized on a periodic spatial grid with spacing Δx and timestep Δt, where u_i^{t+1} depends only on the snapshot u^t.

A.2.1 Lax-Friedrichs Scheme

We use the explicit one-step Lax-Friedrichs scheme [Lax, 1954]:

$$u_i^{t+1} = \frac{1}{2}\left(u_{i-1}^t + u_{i+1}^t\right) - \frac{\nu}{2}\left(u_{i+1}^t - u_{i-1}^t\right), \tag{7}$$

where $\nu = v \Delta t / \Delta x$. Rearranging gives

$$u_i^{t+1} = \frac{1+\nu}{2}\, u_{i-1}^t + 0 \cdot u_i^t + \frac{1-\nu}{2}\, u_{i+1}^t. \tag{8}$$

For $|\nu| \leq 1$, the two neighbor coefficients are nonnegative and sum to one [Courant et al., 1967]. Unlike the heat update, the center coefficient is zero, so the next state uses only neighboring values.

A.3 Reference Fourier Symbols

For a periodic stencil with source-offset coefficients $s(d)$, its Discrete Fourier Transform multiplier under our offset convention is

$$\widehat{S}(k) = \sum_d s(d) \exp\left(\frac{2\pi \mathrm{i} k d}{N_x}\right), \qquad \theta_k = \frac{2\pi k}{N_x}. \tag{9}$$

For the heat stencil $[r, 1-2r, r]$, this gives

$$\widehat{S}_{\mathrm{heat}}(k) = 1 - 2r\,(1 - \cos\theta_k)\,. \tag{10}$$

For the Lax-Friedrichs stencil $\left[\frac{1+\nu}{2}, 0, \frac{1-\nu}{2}\right]$, this gives

$$\widehat{S}_{\mathrm{LF}}(k) = \cos\theta_k - \mathrm{i}\nu \sin\theta_k. \tag{11}$$

The sign convention follows the source-offset definition used in the full-model Fourier analysis and is verified by direct sinusoidal propagation.

B Parameter Selection Details

For both schemes, the dimensionless parameter takes values 0.1, 0.25, and 0.4. The heat equation uses $r = \alpha \Delta t / \Delta x^2$, while Lax-Friedrichs uses $\nu = v \Delta t / \Delta x$. These values satisfy the respective stability conditions $r \leq 1/2$ and $|\nu| \leq 1$.

Table 1: Heat-equation parameter values. The stencil is $[r,\ 1-2r,\ r]$, and its coefficients sum to one.

r	**Stencil**	**Center weight**
0.10	$[0.10,\ 0.80,\ 0.10]$	80%
0.25	$[0.25,\ 0.50,\ 0.25]$	50%
0.40	$[0.40,\ 0.20,\ 0.40]$	20%
0.50 (limit; not used)	$[0.50,\ 0.00,\ 0.50]$	0%

B.1 Heat Equation Parameter Selection

Table 1 lists each value of r, the resulting stencil coefficients, and the center weight.

The three values used in the experiments give center weights of 80%, 50%, and 20%.

B.2 Lax-Friedrichs Parameter Selection

For the Lax-Friedrichs scheme, the Courant number is $\nu = v\Delta t/\Delta x$. Table 2 lists each selected value of ν and the resulting stencil coefficients. Unlike the heat scheme, the center weight is zero for every value of ν, so u_i^t does not contribute directly to the update.

Table 2: Lax-Friedrichs parameter values. The stencil is $\left[\frac{1+\nu}{2},\ 0,\ \frac{1-\nu}{2}\right]$, and its coefficients sum to one.

ν	**Stencil**
0.10	$[0.550,\ 0.000,\ 0.450]$
0.25	$[0.625,\ 0.000,\ 0.375]$
0.40	$[0.700,\ 0.000,\ 0.300]$
1.00 (limit; not used)	$[1.000,\ 0.000,\ 0.000]$

C Attention Collection Details

Pytorch's `TransformerEncoderLayer` does not expose attention weights by default. Because our layers use Pre-LayerNorm (`norm_first=True`), self-attention operates on the layer input after normalization, not on the raw residual stream. Before inference, we register forward hooks on each layer; each hook applies the layer's own `norm1` to the intercepted the layer input and calls `self_attn` directly with `need_weights=True` and `average_attn_weights=False` returning per-head weights of shape (B, N_H, N_x, N_x). Hooks are removed after collection. We average attention weights over $B = 51{,}200$ test samples to obtain a single mean attention matrix per head, which we use for all downstream metrics.

D Epoch Selection

Training length was determined separately for each equation. Convergence rates depended on the equation type rather than the CFL parameter, so we used a single epoch count per equation across all CFL values. We trained until validation MSE stopped improving beyond random variation (see Section D.1 for stopping criteria).

Epoch counts are as follows:

Table 3: Training length by epoch count.

EXPERIMENT	**EPOCHS**
Heat equation (all r)	60
Advection Lax-Friedrich (all r)	65

We first trained 5 seeds per condition and verified that all reached the convergence criterion. Once confirmed, we used that epoch count for all seeds at that CFL value.

D.1 Convergence Criteria

We defined the change in validation loss between the early and late training windows as

$$\Delta\mathcal{L} = \overline{\mathcal{L}}_{\text{early}} - \overline{\mathcal{L}}_{\text{late}}. \tag{12}$$

A run was classified as converged when

$$\Delta\mathcal{L} < 10^{-9}. \tag{13}$$

Negative values of $\Delta\mathcal{L}$ were also classified as converged. These values indicate that the later validation-loss window showed no further improvement and may instead reflect fluctuations after convergence. Averaging the loss within each half-window reduces sensitivity to isolated epoch-level random variation that could affect an endpoint training limit.

We selected training durations so that every seed within each PDE family satisfied this criterion: 60 epochs for the heat equation, and 65 epochs for Lax-Friedrichs advection (Table 3).

E Initial conditions

Each trajectory begins from a random superposition of $K = 5$ Fourier modes:

$$s(x) = \sum_{k=1}^{K} a_k \sin(2\pi k x + \phi_k), \qquad a_k \sim \mathcal{U}(-1, 1), \qquad \phi_k \sim \mathcal{U}(0, 2\pi), \tag{14}$$

where a_k and ϕ_k are drawn independently for each trajectory. We normalize the initial field as

$$u(x, 0) = \frac{s(x)}{\|s\|_\infty}, \qquad \|s\|_\infty = \max_x |s(x)|. \tag{15}$$

We use the same procedure and grid for both equations. Because the numerical updates are linear, they change the amplitudes of the initial modes but do not introduce additional frequencies. Modes $k = 1, \ldots, 5$ are therefore present in the generated states, whereas modes $k \geq 6$ are unseen in the data.

F Relative-Offset Attention Metrics

For the held-out mean attention matrix A of one layer and head, define

$$p(d) = \frac{1}{N_x} \sum_{i=0}^{N_x-1} A_{i,(i+d) \bmod N_x}, \tag{16}$$

where d ranges over the N_x signed periodic offsets. Because every row of A sums to one, $p(d) \geq 0$ and $\sum_d p(d) = 1$.

We quantify concentration across relative offsets using normalized Shannon entropy [Shannon, 1948],

$$H_{\text{offset}}(p) = -\frac{\sum_d p(d) \log p(d)}{\log N_x}, \tag{17}$$

with $0 \log 0 = 0$. This quantity is computed over all offsets and ranges from zero for attention concentrated at one offset to one for uniform attention.

Three-point locality is the attention mass assigned to the offsets used by the three-point numerical schemes:

$$L_3(p) = \sum_{d \in \{-1,0,1\}} p(d). \tag{18}$$

For the wider comparison window $W = \{d : |d| \leq 4\}$, retained attention mass is

$$M_W(p) = \sum_{d \in W} p(d), \tag{19}$$

and the normalized weights within that window are

$$p_W(d) = \frac{p(d)}{M_W(p)}, \qquad d \in W. \tag{20}$$

We report $M_W(p)$ because agreement within W could otherwise involve only a small fraction of a head's total attention.

For the model-level trained–random comparison, mean row entropy is

$$H_{\mathrm{row}}(A) = -\frac{1}{N_x} \sum_{i=0}^{N_x-1} \sum_{j=0}^{N_x-1} A_{ij} \log A_{ij}. \tag{21}$$

Attention probabilities below 10^{-10} are clipped when evaluating this quantity. Thus, $H_{\mathrm{row}}(A)$ measures the average concentration of individual attention rows, whereas $H_{\mathrm{offset}}(p)$ measures concentration after averaging by relative offset.

Let

$$\mathbf{a} = \big(p(-1), p(0), p(1)\big), \qquad \mathbf{s} = \big(s(-1), s(0), s(1)\big), \tag{22}$$

where $\mathbf{s}$ contains the three coefficients of the corresponding numerical stencil. Signed stencil correlation is

$$\rho_s(p) = \mathrm{corr}(\mathbf{a}, \mathbf{s}), \tag{23}$$

using Pearson correlation [Pearson, 1895]; absolute stencil correlation is $|\rho_s(p)|$. The implementation returns zero when either three-entry vector has standard deviation below 10^{-10}. Because Pearson correlation is invariant to shifts and positive rescaling, it does not test agreement in coefficient magnitude and is interpreted only as a descriptive comparison of relative weighting across the three offsets.

G Random-Initialization Controls and Reference Assignment

Model-level comparison. For each PDE–parameter condition, we compare 20 trained models with 30 random weight initializations of the same architecture. Each model contributes one value per metric after averaging over its 16 heads. For trained values $x_1, \ldots, x_{n_T}$ and random values $y_1, \ldots, y_{n_R}$, Cliff's effect size is

$$\delta = \frac{1}{n_T n_R} \sum_{i=1}^{n_T} \sum_{j=1}^{n_R} \left[\mathbf{1}(x_i > y_j) - \mathbf{1}(x_i < y_j)\right]. \tag{24}$$

Ties contribute zero. We use exact two-sided Mann–Whitney tests and apply Benjamini–Hochberg correction to the six condition-level p-values separately for each metric [Mann and Whitney, 1947, Cliff, 1993, Benjamini and Hochberg, 1995].

Reference distributions. Within $W = \{d : |d| \le 4\}$, we use $p_W(d)$ from Eq. 20. The finite-difference stencil and empirical training-state autocorrelation are restricted to W, converted to absolute values, and normalized to sum to one; denote them by q_S and q_A. For cutoff K, define stencil weights reconstructed from modes $|k| \le K$ by

$$f_K(d) = \Re\big[\mathcal{F}^{-1}\left\{\mathcal{F}[s](k)\mathbf{1}(|k| \le K)\right\}(d)\big], \qquad q_{F,K}(d) = \frac{|f_K(d)|}{\sum_{e \in W} |f_K(e)|}. \tag{25}$$

For distributions a and b, Jensen–Shannon divergence in nats is

$$D_{\mathrm{JS}}(a, b) = \frac{1}{2} D_{\mathrm{KL}}(a \| m) + \frac{1}{2} D_{\mathrm{KL}}(b \| m), \qquad m = \frac{a+b}{2}. \tag{26}$$

For $N_x = 64$, we initially consider $K \in \{1, \ldots, 8\}$ and remove any $q_{F,K}$ satisfying

$$D_{\mathrm{JS}}(q_{F,K}, u_W) < 0.01, \tag{27}$$

where u_W is uniform on W. The surviving cutoffs are $K \in \{5, 6, 7, 8\}$ in all six conditions. We calculate

$$d_S = D_{\mathrm{JS}}(p_W, q_S), \qquad d_A = D_{\mathrm{JS}}(p_W, q_A), \qquad d_F = \min_K D_{\mathrm{JS}}(p_W, q_{F,K}). \tag{28}$$

Table 4: Held-out random-initialization audit. An initialization or head slot is counted once if the event occurs in any of the six conditions.

Outcome	Initializations	Head slots
Passes entropy threshold	$13/30$	$26/480$
Meets any reference criterion	$1/30$	$1/480$
Meets criteria for stencil agreement	$0/30$	$0/480$
Meets Fourier-only criteria	$0/30$	$0/480$

The Fourier reference is selected only when

$$d_F < 0.95 \min(d_S, d_A). \tag{29}$$

Otherwise, the lower-divergence member of the stencil and autocorrelation references is selected. An exact tie is resolved in favor of the stencil.

Thresholds from random initialization. Within each condition, let $\mathcal{B}$ denote the random heads used for calibration. The concentration threshold is

$$g = Q_{0.05}\left(\{H_{\text{offset}}(p^{(b)}) : b \in \mathcal{B}\}\right). \tag{30}$$

For each reference $r \in \{S, A, F\}$, let $d_r^{(b)}$ be the divergence of random head b. We calculate

$$m_r = \text{median}_{b\in\mathcal{B}}\, d_r^{(b)}, \tag{31}$$

$$\widehat{\sigma}_r = 1.4826\ \text{median}_{b\in\mathcal{B}} \left| d_r^{(b)} - m_r \right|, \tag{32}$$

$$z_r = \frac{d_r - m_r}{\widehat{\sigma}_r}. \tag{33}$$

Where $\widehat{\sigma}_r$ is from [**?**]. If $\widehat{\sigma}_r \le 0$, the implementation uses the population standard deviation; if that is also zero, it uses one. The reference-specific threshold is

$$f_r = Q_{0.05}\left(\left\{ \frac{d_r^{(b)} - m_r}{\widehat{\sigma}_r} : b \in \mathcal{B} \right\}\right). \tag{34}$$

After selecting $r^\star$, an assignment is reported only when

$$H_{\text{offset}}(p) < g \qquad \text{and} \qquad z_{r^\star} < f_{r^\star}. \tag{35}$$

Autocorrelation and Fourier assignments are reported jointly when

$$\min_K D_{\text{JS}}(q_A, q_{F,K}) < 0.01. \tag{36}$$

This combines all three heat conditions and Lax–Friedrichs $\nu = 0.10$.

Held-out random-initialization audit. For each condition, we remove all 16 heads from one random initialization, estimate all thresholds from the other 29 initializations (464 heads), and apply the procedure to the held-out model. We repeat this for all 30 initializations.

The generator resets the same seeds $0, \ldots, 29$ in each condition. The 180 condition rows are therefore evaluations of 30 shared weight initializations across six conditions, not 180 independent initializations. Across conditions, there are 480 unique $(\text{seed}, \text{layer}, \text{head})$ slots and 2,880 condition–head evaluations.

Across the 2,880 condition–head evaluations, 151 pass the entropy threshold and six also meet a reference-specific divergence criterion. All six arise from seed 25, layer 0, head 0: two meet the autocorrelation criterion and four meet the joint autocorrelation/Fourier criterion. Among trained models, 1,739 of 1,920 heads pass the entropy threshold and 94 also meet a reference-specific divergence criterion. Ninety-three meet the joint autocorrelation/Fourier criterion, one meets the autocorrelation criterion, and none meets the criteria for stencil agreement or for Fourier alone. These head counts are descriptive; the trained–random tests use complete models as the independent units.

Table 5: Analytical and measured parameter-tracking quantities. Measured values average all 20 checkpoints and 16 heads per checkpoint.

PDE	Parameter	Analytical	Measured	Mass in $\lvert d\rvert \le 4$
Heat	$r = .10$	.800	.335102	.124795
Heat	$r = .25$	.500	.334708	.104477
Heat	$r = .40$	.200	.334992	.083528
LF	$\nu = .10$	$-.100$	.008095	.129176
LF	$\nu = .25$	$-.250$	.020679	.153228
LF	$\nu = .40$	$-.400$	.143198	.193848

Additional trained–random results. Mean row entropy is lower after training in all six conditions, with complete separation between trained and random models in every condition (Cliff's $\delta = -1$). All six differences remain significant after Benjamini–Hochberg correction. Absolute stencil correlation increases after training in the three heat conditions but decreases in all three Lax–Friedrichs conditions. Three-point locality is lower after training in five of the six conditions.

Of the 1,920 trained heads, 1,739 pass the entropy threshold and 94 also pass the reference-specific divergence threshold. Ninety-three receive the joint autocorrelation/Fourier designation, one receives the autocorrelation designation, and none meets the criteria for agreement with the numerical stencil. Across conditions, the median attention mass within $|d| \le 4$ ranges from 0.084 to 0.194. These results therefore describe relative weighting within a window that contains a limited fraction of total attention and are treated descriptively.

Parameter Tracking

H Parameter Tracking

Let $p(d)$ denote the relative-offset attention weights for one head. Heat centrality and Lax–Friedrichs signed asymmetry are

$$C(p) = \frac{p(0)}{p(-1) + p(0) + p(1)}, \qquad A(p) = \frac{p(1) - p(-1)}{p(-1) + p(1)}. \tag{37}$$

The common normalization used to obtain $p_W(d)$ cancels in these ratios. For the heat stencil $[r, 1-2r, r]$,

$$C_{\text{heat}}(r) = 1 - 2r, \tag{38}$$

giving 0.8, 0.5, and 0.2 for $r = 0.1$, 0.25, and 0.4. Under the paper's source-offset convention, the Lax–Friedrichs stencil is

$$\left[\frac{1+\nu}{2}, 0, \frac{1-\nu}{2}\right], \tag{39}$$

and therefore

$$A_{\text{LF}}(\nu) = -\nu. \tag{40}$$

Each condition contains 20 trained checkpoints with 16 heads per checkpoint. We calculate C or A for every head and fit

$$Y_{cs\ell h} = \alpha + \beta X_c + \varepsilon_{cs\ell h}, \tag{41}$$

where $X_c = 1 - 2r_c$ and $Y = C$ for heat, while $X_c = -\nu_c$ and $Y = A$ for Lax–Friedrichs. The point estimate uses all $3 \times 20 \times 16 = 960$ head rows per PDE arm. All heads are included; the calculation is not restricted by the entropy criterion. Exact agreement with the analytical coefficients corresponds to $\alpha = 0$ and $\beta = 1$.

For each of 2,000 bootstrap replicates, we sample 20 model seeds with replacement separately within each parameter condition, retain all 16 heads from every selected checkpoint, and refit Eq. 41. The reported 95% intervals are the 2.5th and 97.5th percentiles [Field and Welsh, 2007]. The implementation resamples independently within each condition rather than jointly resampling matching numeric seed IDs across parameter values. The intervals are therefore condition-stratified, not paired across parameter values.

Table 6: Parameter-tracking regressions with 95% condition-stratified model-seed bootstrap intervals.

Regression term	Estimate	95% interval
Heat intercept α_C	.334843	$[.332250, .337419]$
Heat slope β_C	.000183	$[-.003916, .004355]$
LF intercept α_A	$-.055262$	$[-.121357, .016326]$
LF slope β_A	$-.450343$	$[-.682254, -.216835]$

Heat centrality remains near one third as its analytical value decreases from 0.8 to 0.2. For Lax–Friedrichs, the analytical asymmetry becomes more negative as ν increases, whereas the measured asymmetry becomes more positive. The quantities C and A describe relative allocation among the three stencil offsets, not the total attention placed there. No minimum retained-mass threshold is applied. Because only three parameter values are available, each regression has one between-condition residual degree of freedom; multi-reference regressions are not used for the workshop claims.

I Attention Substitution Details

For each checkpoint, we average post-softmax attention over 2,048 validation transitions separately for every layer and head:

$$\overline{A}_{\ell h} = \frac{1}{2048} \sum_{n=1}^{2048} A_{\ell h}(u_n^t). \tag{42}$$

We evaluate the substitutions on 2,048 transitions from a distinct test file. The validation and test paths are required to differ, and neither is replaced by the training file when unavailable.

We apply three substitutions simultaneously in all four layers:

$$\text{head-specific:} \quad \widetilde{A}_{\ell h} = \overline{A}_{\ell h}, \tag{43}$$

$$\text{layer-shared:} \quad \widetilde{A}_{\ell h} = \frac{1}{4} \sum_{h'=1}^{4} \overline{A}_{\ell h'}, \tag{44}$$

$$\text{analytical stencil:} \quad \widetilde{A}_{\ell h} = S. \tag{45}$$

The first removes variation across inputs while preserving differences among heads. The second also removes within-layer head differences. The third uses the row-stochastic circulant matrix formed from the heat or Lax–Friedrichs coefficients and shares it across all heads and layers.

Each substitution is applied after attention softmax and before multiplication by the value vectors:

$$Z_{\ell h}(u^t) = \widetilde{A}_{\ell h} V_{\ell h}(u^t). \tag{46}$$

The value and output projections, residual paths, normalization operations, and feed-forward blocks remain unchanged. Replacement matrices are checked for finite entries, nonnegativity, expected dimensions, and row sums of one. The largest discrepancy between an interception-only pass and the ordinary forward pass across all 120 checkpoints is 1.79×10^{-7}.

Let $M_{a,s}$ be the test MSE under evaluation condition a for model seed s. We use O for the original model, H for replacement using each head's validation-set mean, L for replacement using one validation-set mean shared by all heads in a layer, and S for replacement using the known finite-difference weights. We report

$$R_{a/b,s} = \frac{M_{a,s}}{M_{b,s}}, \qquad E_{a,s} = \frac{M_{a,s} - M_{O,s}}{\max(M_{\mathrm{copy},s} - M_{O,s}, 10^{-12})}. \tag{47}$$

Here, $M_{\mathrm{copy},s}$ is the test MSE of the copy-current-state baseline, $\widehat{u}^{\,t+1} = u^t$. One unit of $E_{a,s}$ equals the error gap from the original model to the copy-current-state baseline. All ratios and differences are formed within checkpoints using the same test transitions.

For each condition, we resample the 20 model seeds with replacement to obtain 20,000 bootstrap replicates. Ratios are averaged on the log scale, whereas normalized excess errors are averaged

Table 7: Paired attention substitutions. Ratios are geometric means across 20 model seeds; excess-error quantities are arithmetic means. Brackets give 95% model-seed bootstrap intervals.

Condition	$R_{H/O}$	$R_{L/H}$	$R_{S/H}$	$\overline{E}_H$	$\overline{\Delta E}_{L-H}$	$\overline{\Delta E}_{S-H}$
Heat, $r = .10$	26.70 [23.57, 29.82]	2.55 [2.06, 3.18]	111.55 [76.25, 160.10]	.115 [.098, .133]	.175 [.135, .217]	20.082 [12.747, 28.049]
Heat, $r = .25$	62.13 [51.17, 73.66]	1.81 [1.36, 2.55]	122.19 [94.53, 154.51]	.182 [.147, .217]	.115 [.063, .180]	26.300 [19.467, 33.240]
Heat, $r = .40$	80.65 [62.75, 102.39]	1.77 [1.40, 2.28]	135.39 [105.48, 167.37]	.195 [.156, .235]	.129 [.074, .191]	29.445 [21.872, 37.334]
LF, $\nu = .10$	39.01 [27.95, 54.53]	9.61 [6.21, 14.57]	72.46 [55.94, 95.00]	.072 [.046, .107]	.439 [.345, .533]	6.137 [3.197, 9.838]
LF, $\nu = .25$	29.60 [21.73, 44.02]	41.06 [26.11, 58.20]	95.59 [81.41, 114.91]	.029 [.012, .058]	.633 [.563, .696]	2.784 [1.115, 5.414]
LF, $\nu = .40$	22.63 [16.49, 32.54]	75.80 [48.69, 110.59]	142.88 [108.83, 190.71]	.013 [.008, .022]	.743 [.663, .808]	2.251 [1.092, 4.341]

on their original scale. The 95% intervals are the 2.5th and 97.5th percentiles [Field and Welsh, 2007]. Model seeds are the resampling unit; heads and test transitions are not treated as independent observations.

The first column identifies the PDE and parameter value. $R_{H/O} = M_H/M_O$ compares head-specific fixed-attention replacement with the ordinary model, $R_{L/H} = M_L/M_H$ compares layer-shared with head-specific replacement, and $R_{S/H} = M_S/M_H$ compares the known finite-difference replacement with head-specific replacement. $\overline{E}_H$ is the mean excess error for head-specific replacement, normalized by the error gap from the ordinary model to the copy-current-state baseline. The final two columns report the mean paired differences $E_{L,s} - E_{H,s}$ and $E_{S,s} - E_{H,s}$ across model seeds.

J Full-Model Jacobian and Fourier Comparison

Jacobian construction and checks. For checkpoint s, let $f_s : \mathbb{R}^{64} \to \mathbb{R}^{64}$ be the learned one-step map. We compute exact autograd Jacobians at 16 held-out test states and average them:

$$\overline{J}_s = \frac{1}{16} \sum_{b=1}^{16} \frac{\partial f_s(u_b)}{\partial u_b}. \tag{48}$$

The 16 states are selected from 64 held-out one-step pairs used to check the local linear approximation

$$f_{\text{lin}}(u) = f_s(u_0) + \overline{J}_s(u - u_0). \tag{49}$$

Condition-mean R^2 values range from 0.9997 to 1.0000. The mean elementwise variation of the Jacobian across base states, normalized by the mean absolute entry of $\overline{J}_s$, ranges from 0.0044 to 0.0313 across conditions. These checks support a local, state-averaged linear summary near the held-out data; they do not establish one global linear operator.

Circulant projection and the learned update. For $M \in \mathbb{R}^{N_x \times N_x}$, define

$$c_d(M) = \frac{1}{N_x} \sum_{i=0}^{N_x-1} M_{i,(i+d) \bmod N_x}, \qquad [\mathcal{P}(M)]_{ij} = c_{(j-i) \bmod N_x}(M). \tag{50}$$

The operator $\mathcal{P}$ is the Frobenius-orthogonal projection onto circulant matrices. We report

$$\rho(M) = \frac{\|M - \mathcal{P}(M)\|_F}{\|M\|_F}, \qquad \eta(M) = \rho(M)^2. \tag{51}$$

Because the one-step map can be written $f(u) = u + [f(u) - u]$, the Jacobian of the learned update is

$$K = J - I. \tag{52}$$

Since $\mathcal{P}(I) = I$,

$$(J - I) - \mathcal{P}(J - I) = J - \mathcal{P}(J). \tag{53}$$

Subtracting identity therefore leaves the noncirculant numerator unchanged while measuring it relative to the learned update norm rather than the full one-step response.

The small values of $\eta(J - I)$ support approximate shift invariance of the local learned update. They do not establish agreement with the analytical finite-difference coefficients.

Table 8: Circulance of the full one-step Jacobian J and learned update $K = J - I$. Brackets give condition-specific 95% model-seed bootstrap intervals.

Condition	$\rho(J)$	$\rho(J-I)$	$\eta(J-I)$
Heat, $r = .10$	.00048	.0538 [.0488, .0596]	.305%
Heat, $r = .25$	.00086	.0545 [.0510, .0582]	.303%
Heat, $r = .40$	.00120	.0560 [.0506, .0618]	.329%
LF, $\nu = .10$	.00191	.0727 [.0652, .0805]	.559%
LF, $\nu = .25$	.00280	.0733 [.0613, .0870]	.625%
LF, $\nu = .40$	.00350	.0702 [.0627, .0779]	.523%

Fourier convention and analytical symbols. Using the source-offset convention

$$y_i = \sum_d J(d) x_{i+d}, \tag{54}$$

the multiplier of mode $x_i = \exp(2\pi \mathrm{i} k i / N_x)$ is

$$\widehat{J}(k) = \sum_d J(d) \exp\left(\frac{2\pi \mathrm{i} k d}{N_x}\right), \tag{55}$$

where $J(d) = c_d(\overline{J})$ is obtained from the circulant projection. The analytical one-step symbols are

$$\widehat{G}_{\mathrm{heat}}(k) = 1 - 2r + 2r\cos\theta_k = 1 - 4r\sin^2\left(\frac{\theta_k}{2}\right), \tag{56}$$

$$\widehat{G}_{\mathrm{LF}}(k) = \cos\theta_k - \mathrm{i}\nu\sin\theta_k, \qquad \theta_k = \frac{2\pi k}{N_x}. \tag{57}$$

Identity has $\widehat{I}(k) = 1$.

For mode set $\mathcal{K}$ and normalized weights q_k, define the complex error

$$\mathrm{RMSE}_{\mathcal{K}}(\widehat{J}, \widehat{G}; q) = \left[\sum_{k\in\mathcal{K}} q_k \left|\widehat{J}(k) - \widehat{G}(k)\right|^2\right]^{1/2}. \tag{58}$$

For each model seed,

$$\Delta_s = \mathrm{RMSE}_{\mathcal{K}}(\widehat{J}_s, \widehat{I}; q) - \mathrm{RMSE}_{\mathcal{K}}(\widehat{J}_s, \widehat{G}; q). \tag{59}$$

Thus, $\Delta_s > 0$ favors the PDE symbol and $\Delta_s < 0$ favors identity.

Generated frequency support and weighting. An audit of all heat and Lax–Friedrichs training states places the non-DC Fourier power in $k = 1, \ldots, 5$ to floating-point precision; the largest measured leakage outside this range is 5.551×10^{-16}. We therefore use

$$k = 1, \ldots, 5: \text{ generated/training-supported,} \tag{60}$$

$$k = 6, \ldots, 8: \text{ nearby unseen,} \tag{61}$$

$$k = 9, \ldots, 32: \text{ higher-frequency extrapolation.} \tag{62}$$

On $k = 1, \ldots, 5$, equal-mode weighting uses $q_k = 1/5$. Empirical-power weighting uses

$$q_k = \frac{\sum_{s,t} |\widehat{u}_s^t(k)|^2}{\sum_{j=1}^{5} \sum_{s,t} |\widehat{u}_s^t(j)|^2}. \tag{63}$$

Weighting by the empirical input spectrum corresponds to measuring frequency-response error according to where the observed inputs carry power [**?**]. Equal-mode weighting instead gives every generated mode the same influence. Approximately 68% of the supported power lies at $k = 1$ and 17% at $k = 2$. The equal-mode calculation is therefore retained to prevent the higher supported modes from being dominated by their lower empirical power.

For each condition and weighting, Δ_s is calculated separately for each of the 20 trained model seeds. We resample those 20 complete model rows for 20,000 replicates and report percentile 95% intervals.

Table 9: Paired Fourier comparison on the generated modes $k = 1, \ldots, 5$. Positive Δ favors the analytical PDE symbol and negative Δ favors identity. Brackets give condition-specific 95% model-seed bootstrap intervals.

Condition	Empirical-power Δ	Equal-mode Δ
Heat, $r = .10$	.002477 [.002152, .002788]	.004450 [.003552, .005297]
Heat, $r = .25$	.000534 [-.000320, .001375]	-.003544 [-.005904, -.001226]
Heat, $r = .40$	-.002403 [-.003407, -.001343]	-.013864 [-.016572, -.011035]
LF, $\nu = .10$	.003083 [.001422, .004785]	-.011192 [-.015351, -.006828]
LF, $\nu = .25$	.016852 [.014555, .019083]	.002641 [-.003729, .008843]
LF, $\nu = .40$	.037497 [.033573, .041242]	.033237 [.022453, .043553]

Table 10: Paired Fourier comparison outside the generated frequency support. Negative values favor identity. Brackets give condition-specific 95% model-seed bootstrap intervals.

Condition	$k = 6, \ldots, 8$	$k = 9, \ldots, 32$
Heat, $r = .10$	-.031658 [-.032304, -.031029]	-.272119 [-.273157, -.271090]
Heat, $r = .25$	-.089506 [-.090792, -.088205]	-.693287 [-.694946, -.691482]
Heat, $r = .40$	-.150172 [-.152095, -.148090]	-1.116269 [-1.118086, -1.114221]
LF, $\nu = .10$	-.188086 [-.193482, -.182696]	-1.403582 [-1.405917, -1.401043]
LF, $\nu = .25$	-.197256 [-.207886, -.188525]	-1.404092 [-1.409275, -1.399534]
LF, $\nu = .40$	-.231412 [-.245459, -.219178]	-1.428512 [-1.434116, -1.423329]

Frequency bins, held-out states, and real and imaginary components are not treated as independent observations. Intervals are condition-specific rather than simultaneous.

Under empirical-power weighting, the model-seed intervals favor the PDE in four conditions, are inconclusive for heat $r = .25$, and favor identity for heat $r = .40$. Under equal-mode weighting, they favor the PDE in two conditions, are inconclusive for LF $\nu = .25$, and favor identity in three.

All unseen-range intervals are below zero. This is a frequency-extrapolation result, not ordinary test-distribution generalization, because the data generator does not populate these modes.